Proof for Theorems in "Downsizing Diffusion Models for Cardinality Estimation"

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A Theorems in Section 2

THEOREM 2.1. *Let p_0 be a probabilistic distribution and $\{p_t|t \in [0,T]\}$ be a family of distributions derived by perturbing p_0 using the SDE (??) from time 0 to time T. Let $k(t) = e^{\int_0^t \alpha(s)ds}$, and $\sigma^2(t)$ be the solution of the initial value Ordinary Differential Equation (ODE)*

$$d\sigma^2/dt = \beta(t) - 2\alpha(t)\sigma^2(t)$$
$$\sigma^2(0) = 0,$$

then, for each choice of α and β, it would hold that

$$p_{t,\alpha,\beta} = (\tau_{k(t)} \circ p_0) * \varphi_{\sigma^2(t)} \tag{1}$$
$$\nabla \log p_{t,\alpha,\beta}(x) = k(t)\nabla \log p_{\sigma^2(t)k^2(t),0,1}(k(t)x), \tag{2}$$

where τ in $\mathbb{R}^d$ is the scaling operator $(\tau_k \circ f)(x) = k^d f(kx)$ and $\varphi_{\sigma^2(t)}$ in $\mathbb{R}^d$ is the probability density function of $\mathcal{N}(0, \sigma^2(t)I_d)$.

PROOF. It can be easily verified that under perturbation of the stochastic differential equation $dx = -\alpha(t)xdt + \beta(t)dw$, a random variable $x(0)$ which takes the value x_0 with probability 1 would satisfy the distribution $x(t) \sim \mathcal{N}(\frac{x_0}{k(t)}, \sigma^2(t))$ at time t. Therefore, if $x_0 \sim p_0(x)$,$p_t(x)$ would have the expression

$$\begin{aligned} p_{t,\alpha,\beta}(x) &= \int_{\mathbb{R}} (2\pi\sigma^2(t))^{-\frac{d}{2}} e^{-\frac{\left\|x-\frac{y}{k(t)}\right\|_2^2}{2\sigma^2(t)}} p_0(y)dy \\ &= \int_{\mathbb{R}} (2\pi\sigma^2(t))^{-\frac{d}{2}} e^{-\frac{\left\|x-\frac{y}{k(t)}\right\|_2^2}{2\sigma^2(t)}} k^n(t)p_0(y)d(\frac{y}{k(t)}), \end{aligned}$$

hence $p_{t,\alpha,\beta} = (\tau_{k(t)}p_0) * \mathcal{N}(0, \sigma^2(t))$.
As

$$(\tau_\lambda \circ f) * (\tau_\lambda \circ g) = \tau_\lambda \circ (f * g)$$

and

$$\tau_\lambda \circ \mathcal{N}(0, \sigma^2) = \mathcal{N}(0, \frac{\sigma^2}{\lambda^2}),$$

this means that

$$p_{t,\alpha,\beta}(x) = \tau_{k(t)} \circ \left(p_0 * \mathcal{N}(0, k^2(t)\sigma^2(t))\right),$$

and hence $p_{t,\alpha,\beta}(x) = \left(\tau_k \circ p_{\sigma^2 k^2,0,1}\right)(x)$.
Directly differentiating the formula on both sides immediately gives

$$\nabla \log p_{t,\alpha,\beta}(x) = k(t)\nabla \log p_{\sigma^2 k^2,0,1}(k(t)x),$$

finishing the proof. □

THEOREM 2.2. *Let $s_\theta(x,t) : \mathbb{R}^d \times \mathbb{R}^+ \to \mathbb{R}^d$ be a neural network and p_0 be a probability distribution, and $p_{0,\theta}(x)$ to be the marginal distribution of $\tilde{x}(0)$ after $\tilde{x}(T) \sim \mathcal{N}(0, \sigma^2(T)I_d)$ is perturbed by the reverse diffusion SDE*

$$dx = -[\alpha(t)x + \beta^2(t)s_\theta(x,t)]dt + \beta(t)d\hat{w}$$

from time t to time 0. Using the notation of Theorem A, the KL divergence between p_0 and p_θ can be upper bounded by

$$D_{KL}\left(p_0(x), p_{0,\theta}(x)\right) \le D_{KL}\left(p_{T,\alpha,\beta}, \mathcal{N}\left(0, \sigma^2(T)I_d\right)\right) + \frac{1}{2}\int_0^T \mathrm{E}_{p_{t,\alpha,\beta}(x)}[\beta^2(t)\left\|\nabla \log p_{t,\alpha,\beta}(x) - s_\theta(x,t)\right\|_2^2]dt, \tag{3}$$

while the value of $\log p_\theta(x_0)$ can be lower bounded by

$$\log p_{0,\theta}(x_0) \ge \mathrm{E}_{p_{0T}(x|x_0)} \log \varphi_{\sigma^2(T)}(x) + \int_0^T n\alpha(t)dt - \frac{1}{2}\int_0^T \beta^2(t)\cdot \mathrm{E}_{p_{0t}(x|x_0)}[\|\nabla \log p_{0t}(x|x_0) - s_\theta(x,t)\|_2^2 - \|\nabla \log p_{0t}(x|x_0)\|_2^2]dt \tag{4}$$

with the equal sign holding in both inequalities if there exists a probability distribution q such that $s_\theta(x,t) = \nabla \log q_t(x), \forall t \in [0,T]$.

PROOF. Proof for this theorem is elegantly provided by Song, Durkan, Murray and Ermon in [?]. □

B Theorems in Section 3

THEOREM 3.1. *Let p_0 be the weighted average of several positive distributions $\{p_i\}_{i=1}^n$, such that*

$$\{V_i\}_{i=1}^n := \{x|p_i(x) > 0\}_{i=1}^n$$

are open domains in $\mathbb{R}^d$ intersecting only at their boundaries, and each p_i is Lipschitz with constant k_i in the interior (but not necessarily at the boundary) of V_i. Denote $V_0 = \mathbb{R}^d \setminus \cup_{i=1}^n V_i$ for convenience, and let $\{p_t|t \in (0,T]\}$ be the family of distributions derived by perturbing p_0 according to the VE[?] scheme (results for the VP[?] scheme can be immediately derived from Theorem A), then:

1. *$\forall x \in V_i$ such that $B(x,r) \subseteq V_i$,*
$$\nabla \log p_t(x_0) = \nabla \log p_0(x_0) + \delta_1(x_0,t),$$
with $\delta_1(x_0,t)$ an infinitesimal term of order $O(\sigma(t))$.
2. *$\forall x \in \partial V_i \cap \partial V_j$ with at most one of i and j being zero, if $B(x_0,r) \subseteq V_i \cup V_j$, $p_{0,i}(x_0) + p_{0,j}(x_0) > 0$, and inside $B(x_0,r)$,*
$$\mathrm{d}\left(y, T_{x_0}(\partial V_i \cap \partial V_j)\right) \le h\|y - x_0\|_2^2$$
holds for some constant h and all $y \in \partial V_i \cap \partial V_j$, then $\forall \lambda$, $\nabla \log p_t(x_0 + \lambda\sqrt{t}\vec{n})$ would equal
$$\frac{e^{-\frac{\lambda^2}{2}}(p_{0,i}(x_0) - p_{0,j}(x_0))\vec{n}}{2\sigma(t)\left(\Phi(\lambda)p_{0,i}(x_0) + (1 - \Phi(\lambda)p_{0,j}(x_0))\right)} + \delta_\sigma(x_0,\lambda,t),$$
with $\vec{n}$ the unit vector normal to $T_{x_0}(\partial V_i \cap \partial V_j)$, pointing from V_j to V_i, and $\delta_\sigma(x_0,\lambda,t)$ a residual term of order $O(1)$.
3. *$\forall x \in V_0$, suppose $y_0 \in \bar{V}_i$ is the point closest to x_0 such that $y_0 \in \cup_{i=1}^k \bar{V}_i$. If $P(x \in B_r(y)|x \sim p_0) > M_1 r^k, \forall 0 < r < R$ for some constants M_1, k_1, while*
$$\|z - x_0\|_2^2 - \|y_0 - x_0\|_2^2 \ge \min\{M_2, k_2\|z - y_0\|_2^2\}$$

for all $z \in \cup_{i=1}^k V_i$ and some constants M_2, k_2, then

$$\nabla \log p_t(x_0) = \frac{y_0 - x_0}{\sigma^2(t)} + \delta_{\sigma^2}(x_0, t),$$

with $\delta_{\sigma^2}(x_0, t)$ a term of order $o(\frac{1}{\sigma(t)^{1+\epsilon}})$ for all $\epsilon > 0$.

Proof of 3.1(1). With a change of coordinates, we may assume WLOG that $x_0 = 0$ and that $\nabla p_0(x) = ke_1$ is parallel to the first coordinate axis. Now, consider the cube $C_r = \{x| \|x - x_0\|_1 < r\}$ where $r = \frac{R}{d}$, and we would have

$$\nabla \log p_t(x) = \frac{\int_{\mathbb{R}^d} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \cdot \frac{x}{t} dx}{\int_{\mathbb{R}^d} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx}$$

which can be decomposed into

$$\frac{\int_{C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx + \int_{\mathbb{R}^d \setminus C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx}{\int_{C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx + \int_{\mathbb{R}^d \setminus C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx}$$

We shall now denote the term $\int_{C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx$ by a_1, $\int_{\mathbb{R}^d \setminus C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx$ by a_2; $\int_{C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx$ by b_1, $\int_{\mathbb{R}^d \setminus C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx$ by b_2. Now, it suffices to show

- $a_1 = \nabla p_0(x_0) + O(\sqrt{t})$
- $a_2 = o(t^n)$ for all n
- $b_1 = p_0(x_0) + O(\sqrt{t})$
- $b_2 = o(t^n)$ for all n.

The bounds on a_2 and b_2 are easy to verify, as $\int_{\mathbb{R}^d \setminus C_r} p_0(x) dx \leq 1$ and $e^{-\frac{\|x\|_2^2}{2t}} \cdot \frac{x}{t^{n+1}}$ decays uniformly to zero in $\mathbb{R}^d \setminus C_r$ for all n as $t \to 0$. For the estimation of a_1, we have

$$\begin{aligned} a_1 &= \int_{C_r} p_0(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx \\ &= \frac{1}{2} \int_{C_r} \left(p_0(x) - p_0(-x)\right)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx \\ &= \int_{C_r} \left(\langle \nabla p_0(x_0), x\rangle + \frac{1}{2}\left(\epsilon_{p_0}(x) - \epsilon_{p_0}(-x)\right)\right)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx, \end{aligned}$$

where $\epsilon_{p_0}(x) = p_0(x) - p_0(x_0) - \langle \nabla p_0(x_0), x - x_0\rangle$.
Using the fact that $\nabla p_0(x_0) = ke_1$, we know that for $n > 1$,

$$\int_{C_r} \langle \nabla p_0(x_0), x\rangle (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{\langle x, v_n\rangle}{t} dx = 0,$$

and for $n = 1$,

$$\begin{aligned} &\int_{C_r} \langle \nabla p_0(x_0), x\rangle (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{\langle x, v_1\rangle}{t} dx \\ =& \int_{-r}^{r} (2\pi t)^{-\frac{1}{2}} e^{-\frac{x_1^2}{2t}} \frac{kx_1^2}{t} dx_1 \int_{C_r^{d-1}} (2\pi t)^{-\frac{d-1}{2}} e^{-\frac{x_2^2 + \ldots + x_d^2}{2t}} dx_2 \ldots dx_d \\ =& k \int_{-\sqrt{t}^{-1} r}^{\sqrt{t}^{-1} r} (2\pi)^{-\frac{1}{2}} e^{-\frac{x_1^2}{2}} x_1^2 dx \cdot \left(2\Phi\left(\sqrt{t}^{-1} r\right) - 1\right)^{d-1}, \end{aligned}$$

a term which approaches $k = \nabla p_0(x_0)$ exponentially fast as $t \to 0$. Meanwhile, for the residual term, we have

$$\begin{aligned} &\left\| \frac{1}{2} \int_{C_r} \left(\epsilon_{p_0}(x) - \epsilon_{p_0}(-x)\right)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{x}{t} dx \right\|_2 \\ \leq& \int_{C_r} \epsilon_{p_0}(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \frac{\|x\|_2}{t} dx \\ \leq& \int_{\mathbb{R}^d} \frac{h \|x\|_2^3}{t} (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx \\ \leq& \sqrt{t} \int_{\mathbb{R}^d} h \|x\|_2^3 (2\pi)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2}} dx = O(\sqrt{t}), \end{aligned}$$

hence $a_1 = \nabla p_0(x_0) + O(\sqrt{t})$. Finally, for the bound on b_1, we can similarly verify using the symmetry of C_r that

$$\int_{C_r} (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} p_0(x_0) + \langle x - x_0, \nabla p_0(x_0)\rangle dx$$

equals $p_0(x_0) \cdot \left(2\Phi(r\sqrt{t} - 1)\right)^d$, hence $p_0(x_0)$ minus an exponentially decaying term, and that

$$\begin{aligned} &\left| \int_{C_r} (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} \epsilon_{p_0}(x) dx \right| \\ \leq& \int_{C_r} h \|x\|_2^2 (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx \leq ht \end{aligned}$$

hence $b_1 = p_0(x_0) + O(t)$, finishing the proof. ◻

Proof of 3.1(2). We may assume WLOG that $x_0 = 0$ and $\vec{n} = e_1$. Defining C_r and a_1, a_2, b_1, b_2 as in the proof of 3.1(1), we shall continue to verify that

- $a_1 = \frac{e^{-\frac{\lambda^2}{2}} \left(p_i(0) - p_j(0)\right)}{\sqrt{2\pi t}} + O(1)$
- $a_2 = o(t^n)$ for all n
- $b_1 = \Phi(\lambda) p_{0,i}(x_0) + (1 - \Phi(\lambda) p_{0,j}(x_0) + O(\sqrt{t})$
- $b_2 = o(t^n)$ for all n.

As before, the bounds on a_2 and b_2 can be verified using the uniform convergence of $e^{-\frac{\|x\|_2^2}{2t}} \cdot \frac{x}{t^{n+1}}$ in $\mathbb{R}^n \setminus C_r$.To derive the estimations on a_1 and b_1, we further divide C_r into the subsets $V_i^+, V_i^-, V_j^+, V_j^-$, where $V_i^- = V_i \cap C_r \cap \{x | x_1 < 0\}$, $V_i^+ = V_i \cap C_r \cap \{x | x_1 \geq 0\}$, and V_j^-, V_j^+ are similarly defined.
This way, $p_0 \setminus C_r$ can be decomposed as $p_0 = p_{C_r} + \Delta p + \epsilon_p$, where:

$$p_{C_r}(x) = \begin{cases} p_{0,i}(x_0) & x_1 \geq 0 \\ p_{0,j}(x_0) & x_1 < 0 \end{cases} \tag{5}$$

$$\Delta p(x) = \begin{cases} p_{0,i}(x_0) - p_{0,j}(x_0) & x \in V_i^- \\ p_{0,j}(x_0) - p_{0,i}(x_0) & x \in V_j^+ \\ 0 & \text{else} \end{cases} \tag{6}$$

$$\epsilon_p(x) = \begin{cases} p_{0,i}(x) - p_{0,i}(x_0) & x \in V_i \\ p_{0,j}(x) - p_{0,j}(x_0) & x \in V_j \end{cases} \tag{7}$$

Now, for the estimate of p_{C_r}, we can verify with the methods used to prove 3.1(1) that $\int_{C_r} p_{C_r}(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x - \lambda\sqrt{t} e_1\right\|_2^2}{2t}} dx$ equals

$$\left(\Phi(\lambda) p_{0,i}(x_0) + \left(1 - \Phi(\lambda)\right) p_{0,j}(x_0)\right) + o(t),$$

and that $\int_{C_r} p_{C_r}(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} \frac{x}{t} dx$ equals

$$\frac{e^{-\frac{\lambda^2}{2}}(p_i(0)-p_j(0))}{\sqrt{2\pi t}} + o(t).$$

For ϵ_p, we can also verify with the methods used to prove 3.1(1) that

$$\left|\int_{C_r} \epsilon_p(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} dx\right|$$

$$\leq \int_{\mathbb{R}^d} k \left\|x+\lambda\sqrt{t}e_1\right\|_2 (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2}{2t}} dx,$$

hence belonging to $O(\sqrt{t})$, and that

$$\left\|\int_{C_r} \epsilon_p(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} \frac{x-\lambda\sqrt{t}e_1}{t} dx\right\|_2$$

$$\leq k\int_{\mathbb{R}^n} (2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} \frac{\|x\|_2 \cdot \left(\|x\|_2 + \left\|\lambda\sqrt{t}e_1\right\|_2\right)}{t} dx,$$

hence belonging to $O(1)$.

Finally, for the estimation of Δp, defining $C_r^{d-1} = \{(x_2, ..., x_n)\}$, we have

$$\left|\int_{C_r} \Delta p(x)(2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} dx\right|_2$$

$$\leq \int_{C_r^{d-1}} \int_{-h\|y\|_2^2}^{h\|y\|_2^2} |\Delta p(x)| (2\pi t)^{-\frac{1}{2}} e^{-\frac{(x_1-\lambda\sqrt{t})^2}{2t}} dx_1 (2\pi t)^{-\frac{d-1}{2}} e^{-\frac{\|y\|_2^2}{2t}} dy$$

$$\leq \sup_x |\Delta p(x)| \int_{C_r^{d-1}} \frac{2h\|y\|_2^2}{\sqrt{2\pi t}} (2\pi t)^{-\frac{d-1}{2}} e^{-\frac{\|y\|_2^2}{2t}} dy = O(\sqrt{t})$$

and that, similarly,

$$\int_{C_r^{d-1}} \int_{-h\|y\|_2^2}^{h\|y\|_2^2} |\Delta p| (2\pi t)^{-\frac{d}{2}} e^{-\frac{\left\|x-\lambda\sqrt{t}e_1\right\|_2^2}{2t}} \frac{\left\|(x_1, y)-\lambda\sqrt{t}e_1\right\|_2}{t} dx_1 dy$$

$$\leq \sup_x |\Delta p(x)| \int_{C_r^{d-1}} (2\pi t)^{-\frac{d-1}{2}} e^{-\frac{\|y\|_2^2}{2t}} \frac{2h\|y\|_2^2 \left(h\|y\|_2 + \lambda\sqrt{t} + \|y\|_2\right)}{t\sqrt{2\pi t}} dy$$

$$= O(1)$$

Summing all of the above, we conclude that

$$a_1 = \frac{e^{-\frac{\lambda^2}{2}}\left(p_i(0)-p_j(0)\right)}{\sqrt{2\pi t}} + O(1)$$

and that

$$b_1 = \Phi(\lambda)p_{0,i}(x_0) + (1-\Phi(\lambda)p_{0,j}(x_0) + O(\sqrt{t}),$$

finishing the proof. □

Proof of 3.1(3). We shall still assume WLOG that $y_0 = 0$. Define the ball $B_{t,\delta}$ as $\{x|\ \|x-y_0\|_2 < R\sqrt{t}^{1-\delta}\}$, then, $\nabla \log p_t(x_0) + \frac{x_0}{t}$ would equal

$$\frac{\int_{B_{t,\delta}} \frac{p_0(x)}{(2\pi t)^{d/2}} e^{-\frac{\|x-x_0\|_2^2}{2t}} \frac{x}{t} dx + \int_{\mathbb{R}^d\backslash B_{t,\delta}} \frac{p_0(x)}{(2\pi t)^{d/2}} e^{-\frac{\|x-x_0\|_2^2}{2t}} \frac{x}{t} dx}{\int_{B_{t,\delta}} \frac{p_0(x)}{(2\pi t)^{d/2}} e^{-\frac{\|x-x_0\|_2^2}{2t}} dx + \int_{\mathbb{R}^d\backslash B_{t,\delta}} \frac{p_0(x)}{(2\pi t)^{d/2}} e^{-\frac{\|x-x_0\|_2^2}{2t}} dx}.$$

Still denoting the two terms in the numerator as a_1, a_2 and the two terms in the denominator as b_1, b_2, we now seek to prove that

- $a_1 < b_1 R\sqrt{t}^{-(1+\delta)}$
- $a_2 = o(t^n)e^{-\frac{\|x_0\|_2^2}{2t}}$ for all $n > 0$
- $b_1 > Ct^k e^{-\frac{\|x_0\|_2^2}{2t}}$
- $b_2 = o(t^n)e^{-\frac{\|x_0\|_2^2}{2t}}$ for all $n > 0$

As $e^{-\frac{\|x-x_0\|_2^2}{2t}} \leq e^{-\frac{\|x_0\|_2^2}{2t}} \max\{e^{-\frac{k_2 t^{1-\delta}R^2}{2t}}, e^{-\frac{M_2}{2t}}\}$ on $\mathbb{R}\backslash B_{t,\delta}$, the bound for b_2 is easy to derive. The bound for a_2 can be similarly proven, as for small enough t, $\|x-x_0\|_2 > \|x\|_2 - \|x_0\|_2$ ensures that

$$e^{-\frac{\|x-x_0\|_2^2}{2t}} \|x\|_2 \leq \max\{e^{-\frac{\|x-x_0\|_2^2}{2t}}, e^{-\frac{\|x_0\|_2^2}{2t}}\} \cdot 2\|x_0\|_2.$$

To prove the lower bound on b_1, consider the ball $B_{t^2,0}$. Noting that

$$e^{-\frac{\|x-x_0\|_2^2}{2t}} > e^{-\frac{\|x_0\|_2^2}{2t}} e^{-\|x_0\|_2} e^{-\frac{t}{2}}$$

holds for all $x \in B_{t^2,0}$, $P(x \in B_r(y)|x \sim p_0) > M_1 r^{k_1}$ implies that

$$\int_{B_{t^2,0}} (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x-x_0\|_2^2}{2t}} p_0(x)dx \geq M_1 t^{k_1} \cdot e^{-\frac{\|x_0\|_2^2}{2t}} e^{-\|x_0\|_2} e^{-\frac{t}{2}}.$$

Finally, $a_1 < b_1 R\sqrt{t}^{-(1+\delta)}$ follows immediately from the fact that $\|x\|_2 t^{-1} \leq R\sqrt{t}^{-(1+\delta)}$ in $B_{t,\delta}$. □

C Theorems in Section 4

Theorem 4.1. *Let $\tilde{x}$ be a random variable taking values $\{x_i\}_{1\leq i\leq n}$ with equal probability (as is* ***always*** *the case when we construct p_t from i.i.d. sample points), such that $\frac{1}{n}\sum_{i=1}^n x_i = 0$. Take $p_t(x)$ to be the marginal distribution of $x(t)$ when $x(0) = \tilde{x}$ is perturbed by the diffusion SDE*

$$dx = -\alpha(t)xdt + \beta(t)dw$$

from time 0 to time t. Then, we would have

(1) $E_{p_t(x)} \left\|\nabla \log p_t(x) + \frac{x}{\sigma^2(t)}\right\|_2^2 \leq \frac{f(t,\tilde{x})}{\sigma^4(t)k^2(t)}$.
(2) $E_{p_t(x)} \left\|\nabla \log p_t(x) + \frac{x}{\sigma^2(t)}\right\|_2^2 \leq \frac{f(t,\tilde{x})g(t,\tilde{x})}{\sigma^4(t)k^2(t)}$.

where

$$f(t,\tilde{x}) = \frac{1}{n}\sum_{i=1}^n \|x_i\|_2^2$$

and

$$g(t,\tilde{x}) = \frac{1}{n}\sum_{i=1}^n (e^{\frac{2\|x_i\|_2^2 + f(t,\tilde{x})}{2\sigma^2(t)k^2(t)}} - 1).$$

Proof of 4.1(1). We only need to prove the lemma under the VE perturbation scheme ($\alpha(t) = 0, \beta(t) = 1$), and all other cases follow immediately from the equivalence of perturbation schemes. In this case, $p_t(x)$ can be expressed as

$$p_t(x) = \frac{1}{n}\sum_{i=1}^n \mathcal{N}(x_i, t),$$

meaning that $E_{p_t(x)} \left\|\nabla \log p_t(x) + \frac{x}{\sigma^2(t)}\right\|_2^2$ would equal

$$\int_{\mathbb{R}^d} \frac{\left\|\frac{1}{n}\sum_{i=1}^n \mathcal{N}(x_i,t)(x) \cdot \frac{x_i}{t}\right\|_2^2}{\frac{1}{n}\sum_{i=1}^n \mathcal{N}(x_i,t)(x)} dx,$$

which, by Jensen's Inequality, is no greater than

$$\int_{\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^n \left(\mathcal{N}(x_i,t))(x) \left\|\frac{x_i}{t}\right\|_2^2\right) dx = f(t,\tilde{x}),$$

noting that $k(t) = 1$ under the VE scheme. ◻

Proof of 4.1(2). As $\frac{1}{n}\sum_{i=1}^{n} x_i = 0$,

$$\nabla \log p_t(x) + \frac{x}{\sigma^2(t)} = \frac{\frac{1}{n}\sum_{i=1}^{n}\left(\mathcal{N}\left(x_j, \sigma^2(t)\right)(x) - k\right)\frac{x_i}{\sigma^2(t)}}{\frac{1}{n}\sum_{i=1}^{n}\mathcal{N}\left(x_j, \sigma^2(t)\right)(x)}$$

would hold for any $k \in \mathbb{R}$. Thus, $\left\|\nabla \log p_t(x) + \frac{x}{\sigma^2(t)}\right\|_2^2$ would be no greater than

$$\left(\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\mathcal{N}\left(x_i, \sigma^2(t)\right)(x) - k}{\frac{1}{n}\sum_{j=1}^{n}\mathcal{N}\left(x_j, \sigma^2(t)\right)(x)}\right)^2\right)\left(\frac{1}{n}\sum_{i=1}^{n}\left\|\frac{x_i}{\sigma^2(t)}\right\|_2^2\right).$$

In the above formula, the second term is easily calculated as $f(t, \tilde{x})$. To upper bound the first term, we take k to be $\frac{1}{n}\sum_{j=1}^{n}\mathcal{N}\left(x_j, \sigma^2(t)\right)(x)$, after which the original expression can be proven equal to

$$\int_{\mathbb{R}^d}\frac{1}{n}\sum_{i=1}^{n}\frac{\left(\mathcal{N}\left(x_i, \sigma^2(t)\right)(x)\right)^2}{\frac{1}{n}\sum_{i=1}^{n}\mathcal{N}\left(x_j, \sigma^2(t)\right)(x)}dx - 1.$$

Using the mean value inequality, $\frac{1}{n}\sum_{i=1}^{n}\mathcal{N}\left(x_j, \sigma^2(t)\right)(x)$ can be lower bounded by

$$\left(2\pi\sigma^2(t)\right)^{-\frac{d}{2}} e^{-\frac{\|x\|_2^2 - \frac{2}{n}\sum_{j=1}^{n}\langle x_j, x\rangle + \frac{1}{n}\sum_{j=1}^{n}\left\|x_j\right\|_2^2}{2\sigma^2(t)}}.$$

Plugging the above expression into the denominator makes the original expression integrable, resulting in the final estimation $f(t, \tilde{x}), g(t, \tilde{x})$. ◻

Note: Our model, ADC, estimates the moment of transition from $O(\frac{1}{\sigma^4(t)k^2(t)})$ decay to $O(\frac{1}{\sigma^6(t)k^4(t)})$ decay using $\max\|x\|_2^2$ rather than $\mathrm{E}_{p_0(x)}\|x\|_2^2$. This is due to the fact that, for p_0 with non-bounded support, such a transition might be arbitrarily postponed even if $\mathrm{E}_{p_0(x)}\|x\|_2^2$ remains bounded.

For example, let $\tilde{x_N}$ take the value $-\frac{1}{N}$ with probability $\frac{N^2}{N^2+1}$, the value N with probability $\frac{1}{N^2+1}$, and $p_{t,N}(x)$ denote the marginal distribution of $x(t)$ when $x(0) = \tilde{x_N}$ is perturbed using the diffusion SDE above. Then, we would have

$$\lim_{N\to\infty}\sigma^4(t)k^2(t)\mathrm{E}_{p_{t,N}(x)}\left\|\nabla \log p_{t,N}(x) + \frac{x}{\sigma^2(t)}\right\|_2^2 = 1.$$

for all $t > 0$.

Theorem 4.2. *Define* $\mathrm{Var}(x_0)$ *to be the error term's variance upon approximating* $\log p_{\epsilon,\theta}(x_0)$ *with formula (??). Assuming that*

1. *The partition is dense enough that for the same* y*, fluctuations of* $\Delta t_i \delta_{t,space}(x_0, y)$ *for different* t *in* $[t_{i-1}, t_i]$ *is negligible compared to the total integration error,*
2. *The error terms* $\{\delta_{i,time}(x_0)\}_{i=1}^{N}$ *are negligible compared to* $\{\delta_{\tilde{t}_i,space}(x_0, y(\tilde{t}_i))\}_{i=1}^{N}$*, and*
3. *The sequence* $\{y(\tilde{t}_i)\}_{i=1}^{N}$ *consists of i.i.d. samples from* $\mathrm{U}([0,1]^d)$

hold for all x_0*,* $\mathrm{E}_{x_0\sim p_0}\mathrm{Var}(x_0)$ *would be minimized under a timestep scheme where* $t_i - t_{i-1}$ *is chosen inversely proportionate to*

$$\left(\mathrm{E}_{x_0\sim p_0}[\mathrm{Var}_y \delta_{\tilde{t}_i,space}(x_0, y)]\right)^{\frac{1}{2}}.$$

Proof. The Monte Carlo estimator (6) has bias $\sum_{i=1}^{N}\delta_{i,time}(x_0)$, which is negligible under assumption 2; and variance

$$\sum_{i=1}^{N}(\Delta t_i)^2 \mathrm{Var}_{y(\tilde{t}_i)}(\delta_{\tilde{t}_i,space}(x_0, y(\tilde{t}_i))),$$

which, under assumption 1, can be approximated by

$$\int_0^T h(t)\mathrm{Var}_y(\delta_{t,space}(x_0, y))dt,$$

where $h(t) = \sum_{i=1}^{n}\mathcal{X}_{[t_{i-1},t_i]}\Delta t_i$.

For a timestep scheme that partitions $[0, T]$ into N intervals, we always have $\int_0^T \frac{1}{h(t)}dt = N$. Therefore, the calculus of variations shows that

$$\mathrm{E}_{x_0\sim p_0}\int_{T_{trunc}}^{T} h(t)\mathrm{Var}_y(\delta_{t,space}(x_0, y))dt$$

attains its minimum when

$$\frac{1}{h(t)} = C\sqrt{\mathrm{E}_{x_0\sim p_0}\mathrm{Var}_y(\delta_{t,space}(x_0, y))},$$

and hence when the step size is chosen inversely proportionate to the square root of $\mathrm{E}_{x_0\sim p_0}\mathrm{Var}(\delta_{t,space}(x_0, y))$. ◻